Tutoring 2 - European Economic Policy

13-11-2026

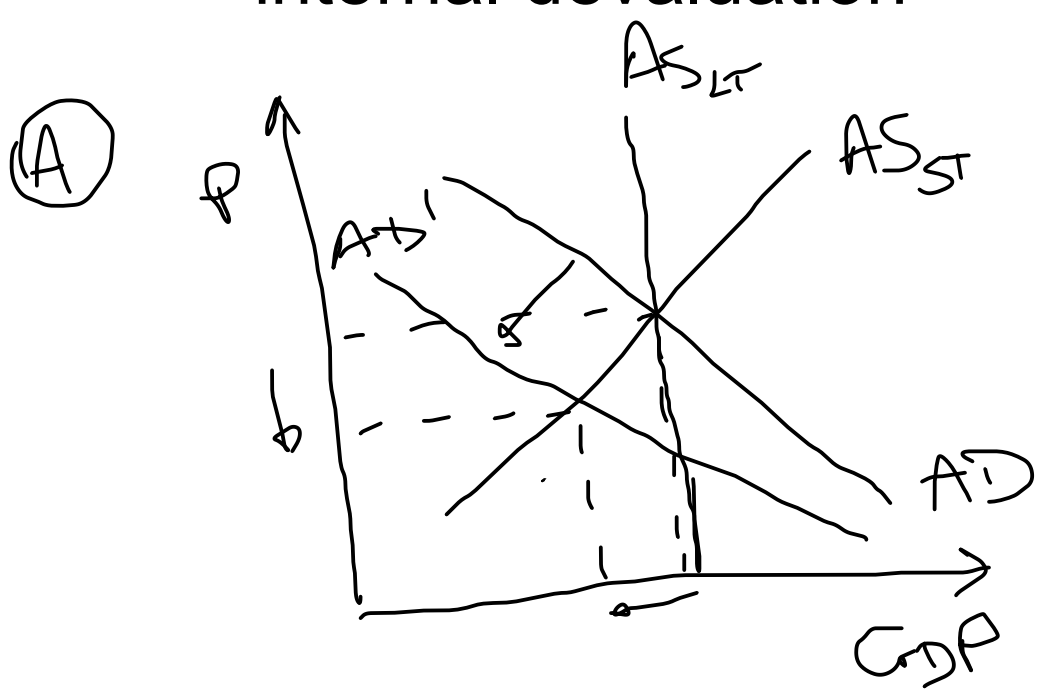
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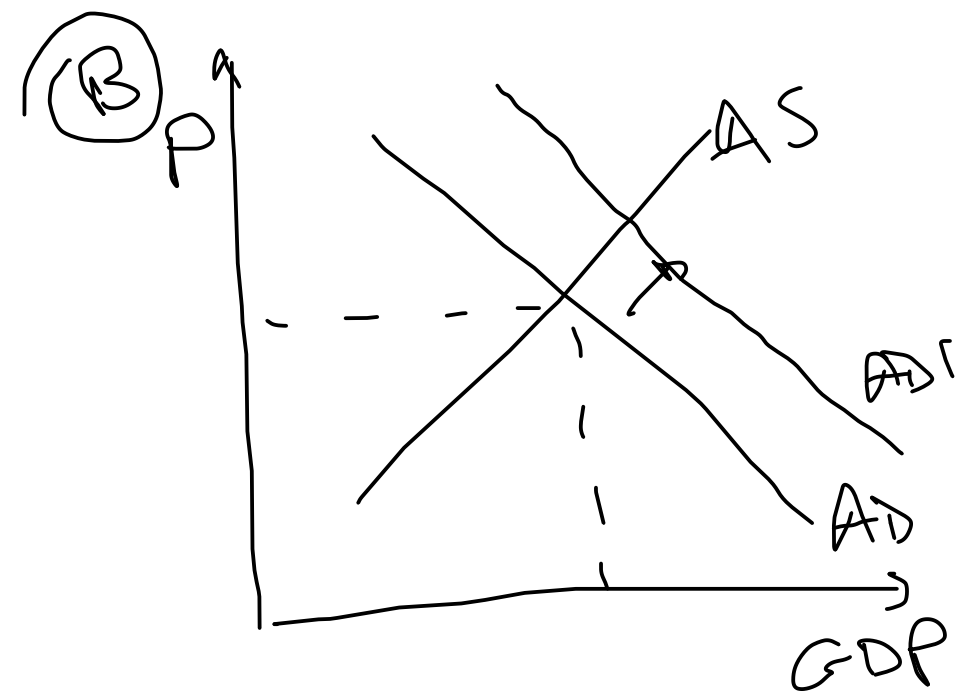
Responses to asymmetric shocks

Let us assume that two independent countries are hit by asymmetric and permanent shock in demand:

- decline in aggregate demand in country A
- increase in aggregate demand in country B

They may resort to national monetary policies to correct the imbalance:
- devaluation
- internal devaluation



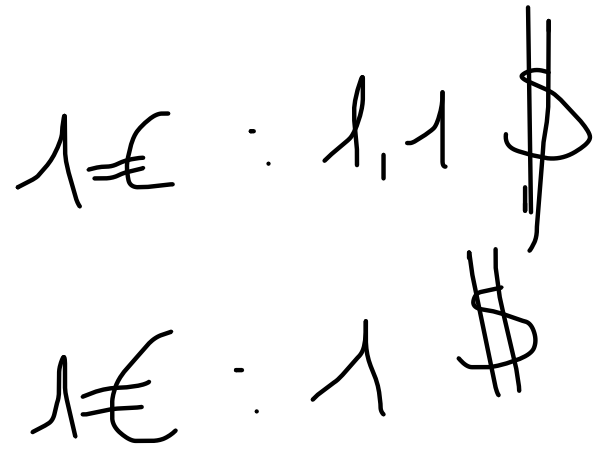


Adjustment through devaluation

Devaluation = manipulation of the nominal exchange rate *E* to re-gain equilibrium after a shock in aggregate demand:

Country A wishes to re-gain competitiveness against country B and depreciates (= devaluates) their national currency.

What are the effects?



Effects of devaluation

1. Given prices, exported goods become cheaper (in country B's currency) and thus more competitive

2. Import prices rises - imported goods become more expensive and less competitive with national goods

3. Consumer prices rise
   Why?

Effects of devaluation on prices

Consumer prices P_C = weighted mean of domestic (P_E) and import prices (P_M)

$P_C = P_E^{1-S} * P_M^S$

We use E to convert import prices to the national currency: $P_C = P_E^{1-S} * (P_M / E)^S$

What happens when E *is* manipulated?
It's easier to estimate the effect if we use the log approximation and take the first derivative wrt time

$$\log(P_C) = \log(P_E^{1-S}) + \log(P_M^S) - \log(E^S)$$

Prop. $\log(x^\alpha) = \alpha \cdot \log(x)$

$$\log(P_C) = (1-S)\log(P_E) + S(\log(P_M) - \log(E))$$

$$\Delta P_C = \Delta P_E^{1-S} \cdot \Delta P_M^S$$

$$\frac{\partial \log(P_C)}{\partial t} \simeq \Delta P_C$$

VARIATION IN PRICE LEVELS OVER TIME

= CONSUMER PRICE INFLATION

$$\frac{\partial \log(P_C)}{\partial t} = (1-S)\frac{\partial \log(P_E)}{\partial t} + S\left(\frac{\partial \log(P_M)}{\partial t} - \frac{\partial \log(E)}{\partial t}\right)$$

$$\dot{P}_C = (1-S)\dot{P}_E + S(\dot{P}_M - \dot{e})$$

$\dot{e} \downarrow 0$

Effects of devaluation on prices

What drives the effect on consumer price inflation?

$log(P_C) = (1\text{-}s)\ log(P_E) + s(log(P_M) - log(E))$

$\dot{p}_C = (1\text{-}s)\ \dot{p}_E + s(\dot{p}_M - \dot{e})$

$\dot{e}$ corresponds to ΔE and $\dot{e}<0$ in case of depreciation

Effects of devaluation on prices

Domestic prices can also rise due to depreciation:

* domestic goods which are produced using imported inputs face an increase in production costs

* indexation (imitation) mechanisms: expected inflation in import-sensitive sectors spills over to other sectors

Effects of devaluation on prices

$\frac{W}{P_C}$ REAL WAGE

* As consumer prices rise due to depreciation of domestic currency, real wage W / P_C declines (due to W not being flexible in the short term)
--> loss in consumers' purchasing power

$P_C \uparrow$

$W \uparrow$ PROD. INPUT

On the long run, wage is renegotiated and adjusted to the rise in price levels

$P_C \uparrow$

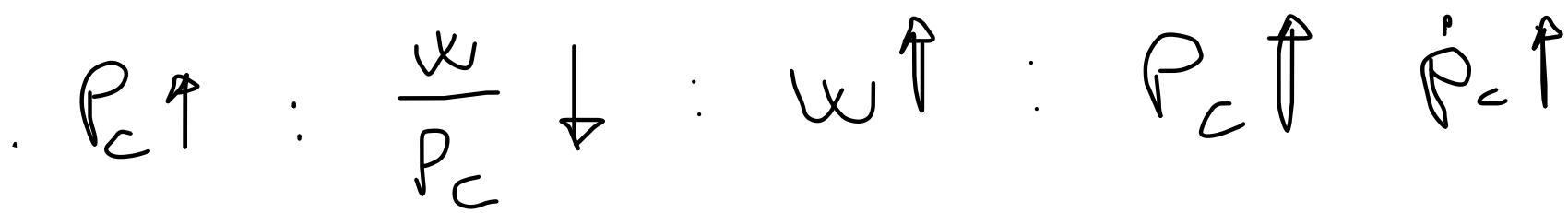
BUT higher wage means higher production costs, which leads to higher prices on the long run

Conclusion on devaluation

After a negative demand shock that reduces GDP and prices, depreciation brings economy back to same level of GDP but with higher prices and lower real wages

Workers negotiates higher nominal wages to re-gain their lost purchasing power
--> higher production costs
--> reduced supply
--> less output produced and higher prices

Manipulation of exchange rate does not increase competitiveness permanently, and increases inflation



Adjustment through internal devaluation

Internal devaluation = reduction of domestic costs
= cut in domestic prices and wages

Country A wishes to re-gain competitiveness against country B but cannot devaluate their domestic currency --> typical in monetary unions, where exchange rate is fixed to $E_R = P_A / P_B$

What are the effects?

Effects of internal devaluation

Two effects on real wage W / P_C

1. Nominal wage W declines (direct effect)

2. Consumption prices P_C decline (due to decline in domestic prices P_E)

What happens to W/P_C?
It declines because the cut in W is higher than the fall in P_C

Conclusion on internal devaluation

After a negative demand shock that reduces GDP and prices, the government cuts wages to gain competitiveness

Since production costs are now lower, supply increases

New equilibrium is reached at previous levels of production, but with lower prices

Unwanted outcome due to decrease in domestic prices:
--> recession
--> deflation

The case of repeated devaluation

We know that depreciation brings economy back to same level of GDP but with higher prices and lower real wages, until workers renegotiates higher nominal wages

Can permanent shocks be corrected by (repeated) exchange rate depreciations?

NOT WORKING

The case of repeated devaluation

* When depreciation becomes recurring, individuals learn to anticipate its effects
* Workers ask for wage increases to prevent the expected decline in their purchasing power W↑ BEFORE PC↑
* Wage increases mean higher production costs
* Prices rise --> inflation rises
* The benefit of competitiveness is lost
* Further rise in prices and inflation may eventually impact on inflation expectations
* Higher inflation expectations are reflected in an **augmented Phillips Curve**

